Tutorial 6 - To Correct Model Inadequacies

In Tutorial 5, we learned that the t-test and F-test are valid only under the MLR assumptions. Otherwise, the resulting inference is no longer reliable. Specifically,

Assumptions. By fitting $y = \beta_0 + \beta_1 x + \varepsilon$, where $\varepsilon \sim \mathrm{N}(0, \sigma^2)$, we believe the data is generated by

(1) independence: the error terms $(\varepsilon_i)_{i=1}^n$ is independent across observations.

(2) normality: the error term $\varepsilon \sim \mathrm{N}(0, \sigma^2)$.

(3) linearity: the true relation is $y \approx \beta_0 + \beta_1 x_1 + \beta_2 x_2$.

(4) constant variance: the error terms variance σ^2 does NOT change with regressors x.

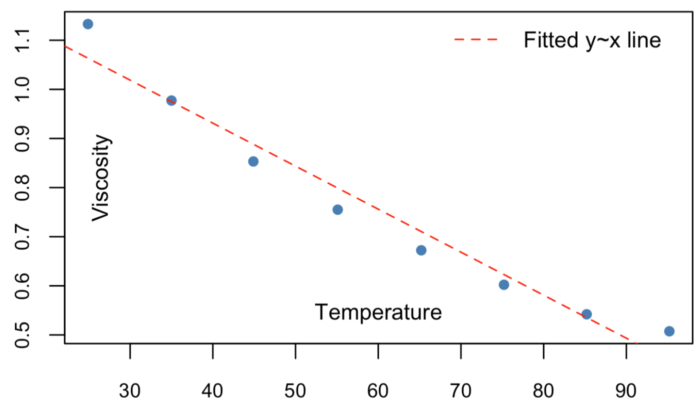
We also knew (2) can be checked by `QQPlot(SR)`, and (3)(4) can be checked by `ScatterPlot(SR vs` $\hat{y}/x$`)`. Our goal in Tutorial 6 is to fix the model when the assumptions are violated. The 3 questions illustrate the following cases:

- Question 1: fixing the non-linearity successful.
- Question 2: fixing the non-linearity failed.
- Question 3: fixing non-constant variance.

Question 1: `viscosity.csv`

(a) `ScatterPlot(visc∼temp)`. Does it like a straight line?

```r
data = read.csv("viscosity.csv")
plot(x = data$temp, y = data$visc)
```



(b) Fit $y \sim x$. Get summary statistics and residual plots. Model adequacy?

```r
model1 = lm(visc ~ temp, data = data)
summary(model1)
```

```
Call: lm(formula = visc ~ temp, data = data)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 1.2815107  0.0468683  27.34  1.58e-07 ***
temp       -0.008758  0.0007284  -12.02  2.01e-05 ***

Residual standard error: 0.04743 on 6 DF
R-squared: 0.9602, Adjusted R-squared: 0.9535
F-stat: 144.6 on 1 and 6 DF, p-value: 2.007e-05
```

A promising summary:

- significant regressors,
- significant F-test,
- satisfactory R^2.

Does it mean we are good to go?

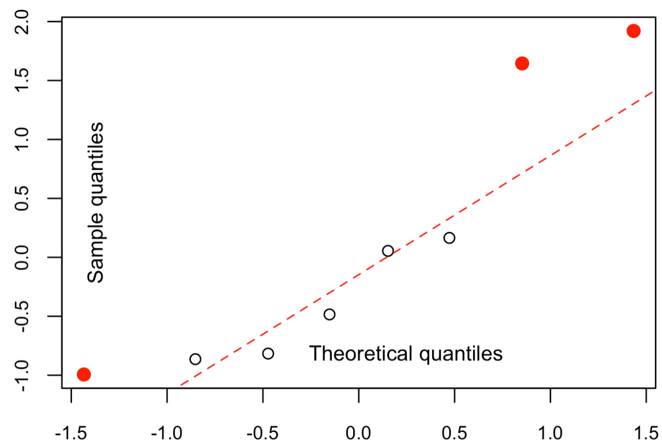
Under model assumptions, especially normality,

$$\mathrm{SR}_i := \frac{e_i - \star}{\square} \approx \mathrm{N}(0,1)^1.$$

So if the distribution/`QQPlot` of $(\mathrm{SR}_i)_{i=1}^n$ deviates from N(0, 1) too much, the normality may be violated.

```r
SR = rstandard(model1)
qqnorm(SR) # QQplot of SR
qqline(SR) # Add ref line for N(0,1)
```

What's your comment on the normality assumption?



Now let's check `Scatter(SR vs` $\hat{y}/x$`)`.

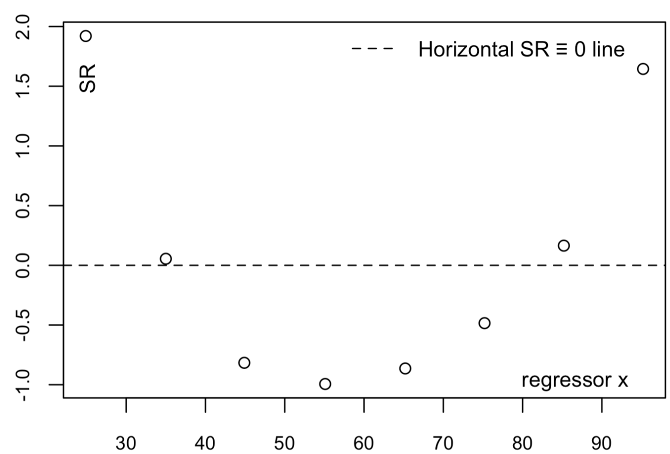
```r
plot(x = data$temp, y = SR)
abline(h = 0) # Add ref line y = 0
```

The plot of SR against x shows a quadratic pattern: $\mathrm{SR} \approx (x-55)^2$. Recall the def of SR above, it means

$$e_i \approx \square(x-55)^2 \quad \Leftrightarrow \quad y_i \approx \hat{y}_i + \square(x-55)^2.$$

Cuz now we are fitting $\hat{y}_i \sim x$, the above suggests

$$y_i \approx \square + \square x + \square(x-55)^2 \quad \text{is a better choice.}$$



Fitting $y \sim x + (x-55)^2$ is equivalent to $y \sim x + x^2$ (just reach out if you wonder why). So we will fit $y \sim x + x^2$ in Q1(c). And the `Scatter(SR vs` $\hat{y}$`)` is left for you to complete.

> **Common Remedy 1** (for non-linearity). When there is violation, we usually start with the most severe one, like the `ScatterPlot(SR vs x)` here. For non-lineariry, we can try adding higher order/interaction terms.

(c) Repeat (b) using proper transformation.

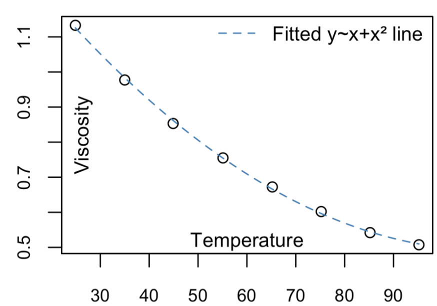
```r
model2 = lm(
    visc ~ temp + I(temp^2),
    data = data))
summary(model2)

# On the most right,
# Draw fitted y~x+x^2 curve
```

```
Call:  lm(formula = visc ~ temp + I(temp^2))

Coefficients:
              Estimate  Std. Error  t value  Pr(>|t|)
(Intercept)   1.55e+00  1.81e-02    85.6     4.1e-09  ***
temp         -1.93e-02  6.62e-04   -29.2     8.8e-07  ***
temp^2        8.81e-05  5.43e-06    16.2     1.6e-05  ***

Residual standard error:  0.00710 on 5 DF
R-squared:  0.999, Adjusted R-squared:  0.999
F-stat:  3360 on 2 and 5 DF, p-value:  1.5e-08
```

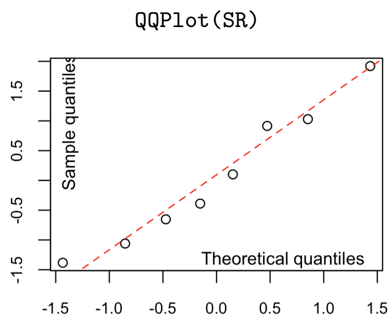


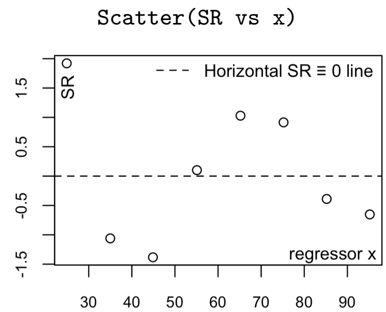
Model 2: $y \sim x + x^2$ has a higher R^2 and a better fitted curve. And its `QQPlot` and `ScatterPlot` no longer indicate serious violations below.

[1] $\star = 0$ and $\square = \sqrt{\hat{\sigma}^2(1-h_{ii})}$ are the mean and se of e. For why they equal these number, see `tut5-script-annotated.pdf`

```r
SR = rstandard(model2)
qqnorm(SR) # QQplot of SR
qqline(SR) # Add ref line for N(0,1)

plot(x = data$temp, y = SR)
abline(h = 0) # Add ref line y = 0
```





Model 2: $y \sim x + x^2$ already works well. And also feel free to try M3/M4 (e.g. $\log y \sim \log x$) in the ref answer, because

> **Common Remedy 2** (for non-linearity). We can apply log on response/regressors.

> **Good News.** Sometimes, after correcting one major violation, another violation may also become less serious or even disappear.

(d) Try Box-Cox transformation. Fit a new model based on it. Any inadequacy?

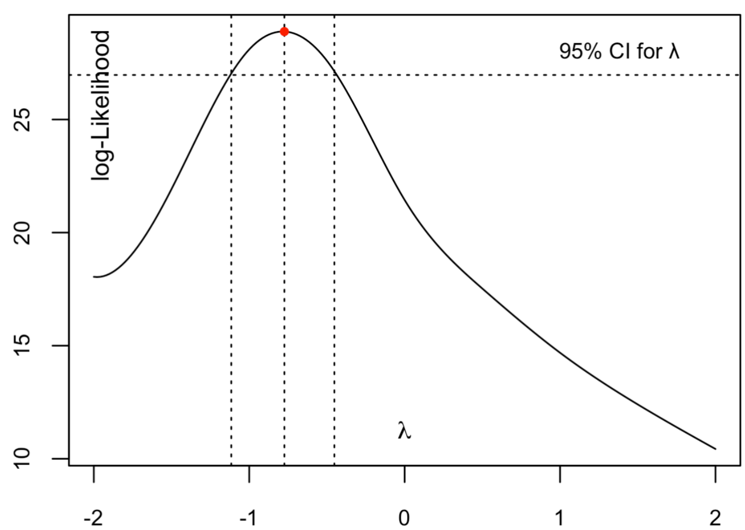
Instead of choosing a transformation manually from the plot, the Box-Cox method gives an automatic way. In short, given the data $(x_i, y_i)_{i=1}^n$, the Box-Cox procedure suggests a value of λ. Based on this value, we transform the response

$$y^{(\lambda)} = f(y, \lambda) \quad \text{looks abstract, but this is some known and fixed function}^2.$$

> **Common Remedy 3** (for non-linearity). We can apply Box-Cox on the response.

```r
library(MASS) # Box-Cox is in it
boxcox(model1,
    # Boxcox on the response of model1
    # Recall model1: y~x
    # So it means fit: f(y ,)~x
       lambda = seq(-2, 2, by = 0.5),
    # Try lambda in grid {-2,-1.5,...,2}
       plotit = TRUE)
    # Return the plot on our right
```

By the `95% CI for` λ, we can choose a convenient value within this interval that has a practical meaning. Here, $\lambda = -1$ is a natural choice, giving $f(y, \lambda) = 1/y$.



Based on the Box-Cox, we may fit `model3: 1/y~x`. The fitting and diagnostic checking are left to you.

> **Common Remedy 4** (for non-linearity). We can apply Box-Tidwell for the regressor.

[5]See `Topic4-To-Correct-Model-Inadequacies.pdf` Page 35 for Box-Cox response and Page 41 for Box-Tidwell regressor.

```r
library(car) # Box-Tidwell is in it
bt = boxTidwell(y ~ x, data = data)
print(bt) # Print the optimal
```

```
MLE of lambda Score Statistic (t) Pr(>|t|)
  -0.038567     33.224   4.644e-07 ***
```

It suggests a new regressor $x^{(\lambda)} = x^{-0.039}$, and we can fit $y \sim x^{(\lambda)}$.

Question 2: `vapor.csv`

This question is left to you. We shall find the violation can not be fixed whatever we try.

> **Bad News.** Unfortunately, not every violation can be corrected. In fact, this is often the case in real data.

Question 3. Consider the SLR model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ where $\mathrm{Var}(\varepsilon_i) = \sigma^2 x_i^2$.

(a1) If the data are truly generated from this (not the constant variance now), will there be any computing issue for R functions `lm(...)/summary(...)`?

Solution. In practice, `lm(...)` and `summary(...)` can still run without any issue, because all following quantities can still be **computed**:

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}} \qquad \hat\beta_0 = \bar y - \hat\beta_1 \bar x \qquad \hat y_i = \hat\beta_0 + \hat\beta_1 x_i \qquad e_i = y_i - \hat y_i$$

$$\hat\sigma^2 = \frac{\sum_{i=1}^n e_i^2}{n-2} \qquad \mathrm{se}(\hat\beta_1) = \sqrt{\frac{\hat\sigma^2}{S_{xx}}} \qquad t = \frac{\hat\beta_1}{\mathrm{se}(\hat\beta_1)}$$

```
Call: lm(formula = y ~ x)

Coefficients:
            Estimate   Std.Err.  t value  Pr(>|t|)
(Intercept)  2.185      0.4126    5.291   0.0016 **
x            0.732      0.1184    6.181   0.0007 ***

Residual standard error: 1.284 on 10 DF
R-squared: 0.7924,  Adj R-squared: 0.7716
F-statistic: 38.2 on 1 and 10 DF
```

Computation is still feasible. However, only under all MLR assumptions, especially $\varepsilon \sim \mathrm{N}(0, \sigma^2)$:

- the $\hat\sigma^2$ is an unbiased estimator for $\mathrm{Var}(\varepsilon) = \sigma^2$.
- the t-value $\sim t_{n-2}$ distribution under H_0.

Now that $\mathrm{Var}(\varepsilon_i) = \sigma^2 x_i^2$:

- the $\hat\sigma^2$ is no longer unbiased.
- the t-value is no longer t_{n-2} (even under H_0).

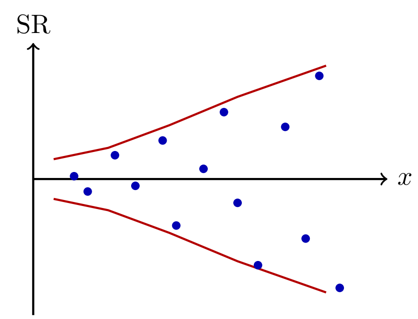
Challenge: How about R^2? Does it lose its meaning?

(a2) What will we see in the `ScatterPlot(SR)`?

Under $\varepsilon \sim \mathrm{N}(0, \sigma^2 x^2)$, the residual follows

$$e_i \sim \mathrm{N}\big(0, \sigma^2 x_i^2 (1 - h_{ii})\big).^6$$

Therefore, the standardized residual ($\mathrm{SR}_i := \frac{e_i - \star}{\square}$) also have a $\mathrm{Var}(\mathrm{SR}_i) \propto x_i^2$. In particular, we expect spread of SR increases with x. So the diagnostic `ScatterPlot(SR vs x)` should display a funnel shape.



`ScatterPlot(SR vs x)`

[6] For the detailed derivation, see `tut5-script-annotated.pdf`.

(a3) Does the new response $y' = y/x$ stablize the variance?

Yes, because the new response has variance

$$\mathrm{Var}(y') = \mathrm{Var}(y/x) = \frac{1}{x^2} \cdot \mathrm{Var} y = \frac{1}{x^2} \cdot (\sigma^2 x^2) = \sigma^2.$$

> **Common Remedy 5** (for non-constant var). When we detect funnel shape in `ScatterPlot(SR vs` $\hat{y}/x$`)`, we can try y/x as new response. (In practice, $\log y$ or Box-Cox may also help).

(b) What is the linearity assumption in the model $y \sim x$? Setting $y' = y/x$ and $x' = 1/x$, how can we express this same assumption in terms of y' and x'? What does it mean?

In the original model, the linearity assumption is

$$y = \beta_0 + \beta_1 x + \varepsilon.$$

Dividing both sides by x, we can equivalently write

$$\frac{y}{x} = \beta_0 \cdot \frac{1}{x} + \frac{\beta_1 x}{x} + \frac{\varepsilon}{x}$$

and after substituting $y' = y/x$, $x' = 1/x$, and $\varepsilon' = \varepsilon/x$, this becomes

$$y' = \beta_0 x' + \beta_1 + \varepsilon'.$$

Therefore, the same linearity assumption can be expressed in the new notation as

$$y' = \beta_1 + \beta_0 x' + \varepsilon'.$$

That is, under this rewriting, the intercept becomes β_1 and the coefficient of x' becomes β_0.